Koa Shen

MATERIALS ENGINEERING | SUMMER 2027

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PROFESSIONAL SUMMARY

Mechanical engineering student with applied experience comparing metals and fiber-reinforced composites for lightweight hardware, supporting a composite seat program, and machining steel and aluminum components. Uses hand calculations, FEA, manufacturing constraints, and hardware evaluation to inform material and design decisions.

EDUCATION

University of California, Santa Barbara | B.S. Mechanical Engineering
Santa Barbara, CA | Expected June 2028 | Major GPA: 3.97 | Cumulative GPA: 3.94
Dean's Honors every quarter | Relevant coursework: Mechanics of Materials, Dynamics, Circuits, MATLAB

MATERIALS AND ENGINEERING EXPERIENCE

Fabrication Team Lead | Gaucho Racing, Formula SAE Electric, UC Santa Barbara
Fall 2024–Present | Santa Barbara, CA

- Secured $2,000 in undergraduate research funding for steering-wheel R&D; used hand calculations and FEA to compare CFRP, GFRP, 6061-T6, and 7075-T6 options for a lightweight baseplate.
- Evaluated the steering wheel for ergonomics, strength, and Formula SAE compliance; iterative development reduced assembly weight 25% and cost 30% versus the prior design.
- Contributed to the team's first composite-seat program, establishing a baseline for material selection, mold and layup process, fit, and installation; finished seat weighed approximately 4 lb.
- CNC-machined 33 suspension parts from 4130 steel and 7075-T6 aluminum, holding critical bearing interfaces within 0.0005 in.

Mechanical Engineering Intern | SLAC National Accelerator Laboratory, LCLS
Summer 2026–Present | Menlo Park, CA; extended part-time remote

- Designed optics mounts, alignment features, and detector strain relief in Siemens Solid Edge; applied GD&T and Teamcenter PDM documentation for manufacture and inspection.
- Performed statics and dynamics hand calculations to evaluate stages and mounts against loading cases protecting detectors valued at more than $100,000.
- Built a Python/Drake digital twin for a 30-DOF, 305-body assembly; collision checks identified at least five pinch points to guide hardware redesigns.

Maintenance Technician | Exploratorium
Summer 2025 | San Francisco, CA

- Reverse-engineered, machined, and welded replacement parts for public exhibits; redesigned a load-spreading bezel to address acrylic cracking and avoid $3,000+ in replacement costs.
- Redesigned encoder mounts for wear resistance and serviceability and documented CAD and remanufacturing instructions for future technicians.

SELECTED MATERIALS AND DESIGN PROJECT

Lightweight FSAE Steering Wheel | Gaucho Racing

- Compared composite and metallic baseplate materials using structural calculations and FEA, then validated the selected steering-wheel design for ergonomics, strength, and competition rules.
- Integrated buttons, potentiometers, display, and quick-release hardware through iterative design and vehicle integration.

TECHNICAL SKILLS

Materials: CFRP, GFRP, 4130 steel, 6061-T6 aluminum, 7075-T6 aluminum, PETG, ABS; comparative material selection
Engineering: Mechanics of materials, hand calculations, FEA, GD&T, SolidWorks, Solid Edge, DFM, Teamcenter PDM
Fabrication and Analysis: Composite design exposure, CNC and manual machining, welding, 3D printing, Python, MATLAB